Design of a Dual-Device Wearable System for Silent Morse-Based Communication with Health-Aware Emergency Response

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***Abstract*—In certain environments, silent yet effective modes of communication become crucial because traditional modes of communication are either infeasible or impractical, such as speech and visual cues. The methodology involved in designing and implementing a two-way wearable system capable of communicating through Morse code is provided herein. According to the described methodology, in the proposed design, the system consists of a transmitting wristband device called TX and a receiving smartwatch device referred to as RX, and data exchange is achieved via the BLE channel. The gestures of the user, such as the movement to the right for a dot and to the left for a dash, are detected by the IMU sensor in the TX wristband and converted to Morse code. and is decoded by the RX smartwatch with the help of an error-correction machine learning module. The above method was used in developing this system to solve problems that occur during traditional communication methods in such environments where noise is inevitable, as well as solving problems that occur when dealing with people with disabilities, in which case they are not able to communicate verbally since this method involves sending out messages in Morse code without making any noise. Additionally, feedback mechanisms like display and haptics have been integrated. A special feature implemented in the TX wristband involves the capability of monitoring health status; if an abnormality is detected, then an alert message (SOS) is sent by the TX wristband automatically. This system will offer reliable, flexible, and user-controlled communication without the requirement of predefined gestures or button actions, thus making communication more effective and efficient even without sound.**

Keywords— wearable communication, Morse code, Bluetooth Low Energy, IMU sensor, machine learning, health monitoring

I. Introduction

Smart wearables, including but not limited to smart watches (e.g., Apple Watch, Samsung Galaxy watch), and fitness wristbands (e.g., Mi band, Fitbit) have emerged as key players in the modern world in recent years because of the benefits of compactness, portability, and the ability to offer real-time interactive services. These developments have been possible owing to advancements made in the field of embedded systems and sensors technology, and such gadgets have been extensively used in applications like healthcare monitoring, fitness management, industrial safety, and communication aid applications [4]– [6]. The global rise of wearable devices shows the potential benefits that such devices can bring to human computer interaction, as well as service provision based on data gathering at all times **[4], [6]**.

However, despite all these developments, there are still many situations where ordinary communication tools such as voice and vision cannot be effectively employed. Situations involving high noise levels such as building sites, industrial plants, disaster areas, and rescue missions are difficult for sound transmission [4]. Moreover, people who suffer from speech or hearing problems find it difficult to communicate through normal modes of communication systems that are silent, efficient, and reliable [1], [2].

Recent studies have proposed gesture recognition-based interaction systems and Morse code-based communication as possible solutions to this problem. Gesture recognition techniques, especially those based on machine learning and deep learning algorithms, have shown significant success in recognizing and assigning commands to pre-defined arm gestures and mapping them to specific commands [3], [7]. This technique has been used for various purposes, such as smart homes, robotics, and wearable-based interaction systems. The limitation of these types of methods is that they require the users to use fixed gesture sets for communication, which limits their application. Another approach that can be used in this regard is the Morse code-based system; it is an efficient technique for communication but rely on button-based input methods [1], [2].

Under these conditions, there is a requirement for a flexible and portable communication device that uses an efficient input method along with a coding method that effectively addresses these issues. Moreover, incorporating other functions such as health monitoring and alarm systems can increase the practicality of these devices [5], [10].

The study proposes a wearable communication system that consists of two components – the TX wristband and the RX smartwatch. With this system, the user can input the Morse code through gesture inputs where tilting right represents a dot and tilting left represents a dash. The incoming data collected from the RX component would be translated using an error-correction algorithm based on machine learning and ensuring accuracy [13],[14]. In addition, this proposed wearable communication system offers real-time feedback in terms of visual display and haptic feedback along with monitoring the health condition of the user automatically initiating emergency alarms under unusual circumstances.

Important considerations for the suggested research project are:

- Construction of a two-device wearable setup that can facilitate silent communication.
- Utilization of gesture detection algorithms to enable Morse code conversion by using IMU sensors.
- Application of machine learning algorithms to enhance error correction.
- Health monitoring and triggering of alarms in case of an emergency.
- Development of an innovative system.

II. RELATED WORKS

Due to the advancements made in recent times regarding wearable technology and man-machine interaction, several alternative communication devices have been developed that rely on signal encoding through gestures. Research has been conducted in the area using IMUs (inertial measurement unit) in wearable technologies to detect motions in the hands and arms [3], [7]. This is accomplished by incorporating accelerometers and gyroscopes to identify certain movements.

The use of gesture-based communication systems, which include machine learning and deep learning methods, is highly accurate at identifying predefined gestures made by the arms. For example, the application of deep learning systems has been applied in classifying arm gestures and mapping them to predefined gestures in search and rescue missions and emergencies [3], [7]. The system is highly accurate and provides a means of communication in situations that require efficiency. However, the system can only recognize predefined gestures, thereby limiting the possibility of free communication.

Apart from gesture-based communication solutions, Morse based communication has also gained widespread attention in the field of assistive technology. These systems enable people to transmit information through relatively easy input devices like switches or buttons that then translate that input into either written or spoken text [1], [2]. Such solutions are quite helpful for persons having speech or movement disabilities. The problem is that existing Morse-based systems use only manual input devices that do not fit well into wearable solutions.

In addition to this, several assistive communication devices have introduced some other features like text prediction and text to speech conversion for improved usage [11], [12]. Although these technologies provide efficient means of interaction, they do not combine various technologies into one such as gesture based, wearable assistive communication[1],[2],[11]. and context awareness technology.

In spite of the advancements in the field of gesture communication devices as well as Morse input devices, there is still a long way to go towards creating an integrated system that takes into consideration all of the factors including the ease of inputting data and wearable aspects, as well as incorporating intelligent processing techniques.

To mitigate such constraints, it is recommended that the suggested system should adopt the dual device-based wearable communication model, which involves gesture-based Morse code inputting and machine learning based decoding [13], [14] along with an emergency warning mechanism based on the health status of users.

III. PROPOSED WORK

The proposed system will be a two-device wearable communication system that allows silent and efficient real-time communication between the users. The system will comprise a transmitting wristband, referred to as TX, and a receiving smartwatch, referred to as RX, which exchange data via Bluetooth Low Energy (BLE). This system will not only provide communication features but also offer assistance in terms of safety and health.

A. System overview

The TX bracelet tracks the wrist movements through a 3-axis accelerometer and gyroscope within the IMU sensor. Such movement data is analysed for directions of tilting and timing to be mapped into Morse codes. Here, a right-hand tilt means a dot (·) while a left-hand tilt means a dash (–). The pause between gestures exceeding 600 ms(approx.) indicates the end of characters while that exceeding 1200 ms(approx.) marks the end of words.

The coded Morse code is then wirelessly transmitted from the TX bracelet to the RX smartwatch via BLE. Once received at the RX smartwatch, Morse code is converted into readable characters via machine learning-based error correction and displayed on the smartwatch screen along with haptic alerts.

B. System architecture

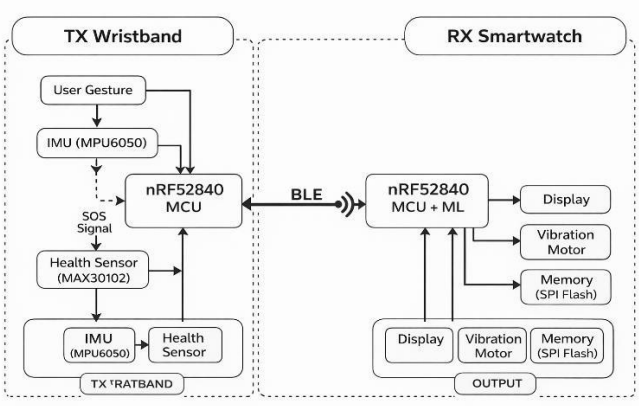


Fig.1. System architecture of the proposed dual wearable communication system showing TX wristband, RX smartwatch, and BLE-based data transmission.

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Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

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C. Gesture Encoding and Timing Mechanism

The recognition of gestures involves checking for thresholds and assessing the trend of IMU measurements with respect to time. When the wrist is tilted by more than ±20° from its resting position, a gesture is detected. Based on the duration of the tilt, a dot or dash is defined as follows:

- Dot: Duration between 150 to 400 ms
- Dash: Duration between 400 to 900 ms

In order to minimize any erroneous gestures due to noise, a debounce time of 100 ms is added after the detection of each gesture. The device continuously samples the IMU readings at around 50 Hz.

D. TX-RX Communication Using Bluetooth

The connection between the TX and RX units is implemented via Bluetooth Low Energy (BLE) due to its ability to consume relatively little energy and ensure reliable transmission of data within close proximity range. The TX unit acts as a BLE peripheral, whereas the RX smartwatch acts as a BLE central unit.

The information coded in Morse is converted into structured packets that are sent in approximately 50-100 ms intervals. A packet consists of gesture type, timestamps, and other parameters. The RX unit receives packets, interprets them, and builds up the corresponding sequence almost instantaneously (in no more than 200 ms).

E. Health Monitoring and Emergency Alerts

TX bracelet constantly monitors your movement and physiological data through its inbuilt sensors and analyses the data for any irregular heartbeats among other factors in Realtime (provided that such a function is present in the device). In case anything goes wrong, the device sends an automatic emergency alert. The message is then sent through Bluetooth Low Energy to the RX smartwatch, initiating a series of continuous vibrations and emergency alerts. If not worn, the user gets an auditory alarm too.

F. Machine Learning Integration into the System

G. In order to increase the reliability of Morse decoding, a machine learning model is incorporated into the RX smartwatch system. It is important to stress that the model serves as an error-correction tool which compensates for inconsistencies and timing variations when translating Morse sequences.

1. Training of the Model
   The training process is implemented offline using the Python programming language. During the process, the machine learning model learns to identify patterns within the Morse sequences based on their timing and spacing. The dataset contains variations in user mistakes and timing, which enhances the performance of the model. The input variables for the machine learning algorithm are the durations of gestures and inter-gestures intervals.

2. Converting and Embedding the Model
   After training, the model is transferred into an appropriate format to be deployed onto the microcontroller. To do this, the model is converted to C-array notation or the TensorFlow Lite for Microcontrollers library. The model is embedded into the firmware of the nRF52840 microcontroller.

3. On-Device Inference:
   In the course of execution, the received Morse codes are decoded in real-time within the RX device. The features extracted from the signal are provided to the ML model for predicting the most likely character or word. The inference process has been kept low-latency and should take only a few milliseconds to perform its function and keep the communication process alive.

4. Integration within the system:
   The ML algorithm is deeply integrated into the process of decoding Morse signals. Once the data is received through BLE, it gets decoded as usual based on mapping rules and then fine-tuned by the ML model, after which the message is presented to the user. This helps improve communication efficiency when users move their hand gestures in an unstructured manner

H. Detailed Hardware Architecture

In order to achieve good performance, the TX unit will be clearly separated from the RX unit, with each optimized according to the respective function required.

1. TX Wristband Hardware Design

TX wristband takes responsibility for gesture recognition, pre-processing, vital sign monitoring, and wireless data transmission. It consists of the nRF52840 microcontroller that is used as a core element for data processing and communication.

MPU6050 IMU sensor is interfaced to the microcontroller through the I2C communication bus. Sensor's SDA and SCL lines are directly linked to corresponding I2C pins of the nRF52840 with pull-up resistors providing reliable communication. IMU constantly collects accelerometer and gyroscope data with sampling rate approximately equal to 50 Hz. Microcontroller analyses tilt direction and duration based on collected data, allowing gestures to be classified as Morse signal codes. The MAX30102 sensor is another device used in TX wristband design and can be interfaced with the microcontroller via the same I2C bus to facilitate multiple devices' connection to one communication line. Sensor constantly monitors heart rate and SpO_2 index. Based on received data, the microcontroller checks abnormal physiological conditions.

The tactile push button is linked to the GPIO pin that has the internal pull-up resistor. On pressing the button, the pin detects a logic low value, giving the user the ability to either start or stop message transfer. Debouncing is achieved in software using delays of about 50 – 100 milliseconds. The haptic motor feedback uses a small coin vibration motor driven by an NPN transistor switch circuit. The transistor's base terminal is linked to the GPIO pin via a resistor, whereas its collector-emitter terminals control the motor supplied with power from the battery. When the GPIO pin is set to a high level, the transistor saturates, enabling current to pass through the motor and produce vibration feedback.

The power source is a lithium-polymer battery that is connected using a JST connector. The battery is charged using a TP4056 module, whereas a voltage regulator (3.3V Low Dropout Regulator) provides constant power to the microcontroller and sensors. Decoupling capacitors are added close to every component. The design utilizes a single compact PCB where the IMU and health sensors are mounted on the bottom side for reliable motion sensing and skin contact, whereas the microcontroller and power circuitry are mounted on the top side.

2. Hardware Design of the RX Smartwatch

The RX smartwatch performs operations such as receiving information from the sender, its processing, and displaying output data to the receiver. The watch contains the nRF52840 microcontroller that serves as the main processor, implementing BLE communication, decoding, and controlling other components.

In order to display data, the TX smartwatch uses an SPI interface between the display (OLED/TFT) and the microcontroller. There are four connections between these devices, namely MOSI, SCLK, CS, and DC.SPI Flash Memory (W25Q32) is also connected to this interface but has a separate chip select line. This memory component stores data in form of messages sent by users. After decoding the received message, the microcontroller writes it to the memory.

A vibration motor operates on a transistorized driver circuit, similar to the one on the TX side. An additional buzzer is connected to the board through a GPIO controlled driver circuit and is useful when a user does not wear the device. Wear detection takes place by means of capacitive sensing pads connected to GPIO or ADC pins. Depending on how much the capacitor value changes, the device will know if it's being worn or not, and based on that, the system can switch between silent vibration alerts and audible buzzer alerts.

BLE communication is taken care of inside the nRF52840 chip, acting as the central node here. Data packet with Morse code is received by the BLE module and decoded in real time. The RX device uses a lithium-polymer battery with a similar design to the one in the TX device in terms of both charging and regulation circuits. Communication channels and power lines are separated from each other on the PCB to ensure proper functionality.

The RX system includes all the necessary components on the same board, with the screen placed at the top and sensing elements placed appropriately.

3. PCB Design Issues:

Both TX and RX units are mounted on specially designed PCBs employing surface mount technology. Important components like MCU, IMU, RAM, and power modules are mounted on the PCB, whereas items like battery packs and vibrational motors are mounted off-PCB with the help of header pins or connectors.
The PCB design is optimized for:
- Portability of wearables
- Effective power transmission
- Interference avoidance
- Heat dissipation

The physical design mounts skin contact sensors like MAX30102 or capacitive pads on the underside of the board, and processor units or interface ports are mounted on the upper side of the board.

I. System Working and Data Flow

The working of the system follows a structured pipeline:

1. Gesture Acquisition:
   The IMU samples wrist movement data continuously at around 50 Hz. If there is a tilt more than a specific threshold value (around ±15°), it is interpreted as gesture input.
2. Morse Coding:
   Depending on how long the gesture takes, it can be categorized as a dot or a dash. Timing values required to determine dots and dashes can be set by the user, and they have an adjustable range (e.g., 150–400 ms for dot, 400–

900 ms for dash). The pauses between gestures allow characters and words to be distinguished.

3. Transmitting Data:
   Packeted encoded Morse messages are sent over BLE connection from TX to RX with low latency (around <200 ms). The TX unit functions as the peripheral device and RX is the central receiving unit.
4. Decoding Messages and Corrections: RX device decodes data sent over the communication channel and uses a machine learning technique to correct for inconsistencies in gesture timings.
5. Displaying Messages and Feedback:
   After decoding the messages, they are displayed on the screen, and vibration feedback is provided to the user. For an emergency situation, constant vibration or audio feedback is activated when detection is successful.
6. Monitoring Health and SOS Functionality: TX device monitors health parameter values of the user constantly.

J. Prototype Testing and Validation

The system is implemented by developing a functional prototype using Arduino development board, MPU6050 sensor, Bluetooth module, and an LCD monitor. It demonstrates the viability of communication through gestures, real-time communication, and message decoding.

Although the prototype utilizes various separate modules to test and validate different aspects of the system design, the actual implementation of the system utilizes integrated circuits on a compact Printed Circuit Board (PCB). The testing and validation outcomes confirm the viability of the system design.

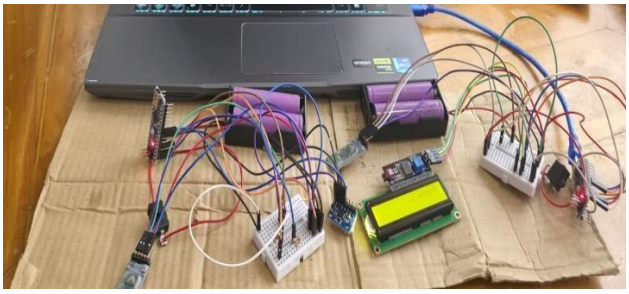


Fig.2. Prototype implementation of the proposed wearable communication system using Arduino, MPU6050, Bluetooth module, and LCD display.

TABLE I.

S.No	Components		
	Component	***Description***	***Quantity***
1.	nRF52840	Microcontroller with BLE	2
2.	MPU6050	IMU (Gesture Detection)	1
3.	MAX30102	Heart Rate & SpO2	1
4.	Display (TFT/OLED)	Message Display	2
5.	SPI Flash Memory	Message Storage	1
6.	Vibration Motor	Haptic Feedback	2
7.	Buzzer	Audio Alert	1
8.	Li-Po Battery	Power supply	2
9.	TP4056 Module	Battery Charging	2
10.	Voltage Regulator	Power Regulation	2
11.	BLE Antenna	Wireless Communication	2
12.	Capacitive Sensor	Wear Detection	2
13.	Push Button	User Input	1
14.	PCB Fabrication	Custom PCB	2

Fig.1. Example of a figure caption. (*figure caption*)

IV. RESULTS

The above-proposed wearable communication system that consists of two devices was tested with a functional prototype using an Arduino microcontroller board, MPU6050 sensor, Bluetooth device, and LCD screen. The prototype helped demonstrate the ability of detecting gestures, sending codes in Morse form, wireless communication, and decoding messages.

A. Accuracy of Gesture Detection Process In terms of gesture detection based on an IMU system, right tilt (dot) and left tilt (dash) motions were detected successfully in all scenarios. The gesture detection system exhibited consistency in gesture recognition despite slight inconsistencies in wrist movements. In addition, unintentional hand movements sometimes led to false detection results, but they were corrected via thresholding and time restrictions.
Using only one axis in the process was convenient.

B. Accuracy of Morse Code Transmission
Morse code sequences generated were transmitted through Bluetooth communication and received as well as decoded by the receiver. Transmission reliability was achieved by the system with a small delay. The timing segmentation used includes:

- Dot length ≈ 200ms
- Dash length ≈ 600ms
- Character length ≈ 600 – 800ms
- Word length ≈ 1200 – 1500ms

The timing values provided effective distinction among Morse symbols leading to accurate decoding.

C. Communication Delay
The average communication delay from TX to RX was found to vary between 100-300ms based on the Bluetooth state. The latency value is insignificant for real-time communication purposes.

D. Health Monitoring and SOS Signal
The health monitoring feature was simulated in order to monitor abnormal body parameters. In case of any anomalies

in the parameters, the system sent an SOS signal automatically from the transmitter wrist band to the receiver smartwatch. SOS feature performed effectively as the automatic generation of alerts was successfully demonstrated.

E. Error Correction Using Machine Learning A simple machine learning error correction system was designed and implemented at the receiving end for improving the accuracy of messages. It helps in correcting any slight errors in the Morse code generated due to inconsistencies in gestures. This will greatly increase the ease of use for new users who do not perform perfect gestures.

F. Prototype Testing

The prototype developed has effectively demonstrated the following features:

Gesture-based real-time Morse code generation
Transmission of data using Bluetooth connection
Decoding and displaying messages on LCD screen
Complete system operation from transmitter to receiver

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